

FINANCIAL LOAN PORTFOLIO ANALYSIS

Professional Data Analytics Project Report

Excel Data Preparation • Power Query • Power BI • SQL Validation

Project component	Implementation
Source / cleaned dataset	Excel workbook
Data cleaning & feature engineering	Excel Power Query
Data modeling & analytical measures	Power BI
Interactive dashboard	Power BI — 4 analytical pages
Independent validation	MySQL SQL
Analytical scope	Portfolio size, recovery, profitability, risk, borrowers, purpose, geography, income and credit

Prepared from the supplied financial loan workbook, PBIX, SQL script and dashboard / Power Query evidence.

1. Executive Summary

This project develops an end-to-end financial loan portfolio analytics solution from a cleaned Excel dataset through Power Query transformation, Power BI modeling and dashboarding, and independent SQL analysis. The cleaned analytical table contains 38,576 loan-level records and 40 fields. The source worksheet contains 24 fields, while the analytical output adds derived measures and classifications for recovery, profitability, risk, income, loan size, term and validation.

At portfolio level, the dataset represents \$435.8M in loan amount and \$473.1M in total payments received. The source logic reports \$71.4M of interest earned from fully recovered loans, producing a portfolio profit-margin measure of 16.4%. Recovery calculated across total payments is 108.6%, while the unrecovered exposure is \$34.1M (7.8% of loan amount).

Executive KPI Snapshot

KPI	Value	Interpretation
Total Loans	38,576	Loan-level records in the cleaned analytical table
Total Loan Amount	\$435.8M	Portfolio amount issued
Total Payments Received	\$473.1M	Aggregate payments recorded
Interest Earned	\$71.4M	Interest contribution from fully recovered loans under project logic
Profit Margin	16.4%	Interest earned ÷ total loan amount
Recovery Rate	108.6%	Total payments ÷ total loan amount
Unrecovered Amount	\$34.1M	Outstanding exposure across non-fully-recovered loans
Charged-Off Rate	13.8%	Share of loans with Charged Off status
High-Risk Loans	7,873 (20.4%)	Loans classified as High risk

Executive Interpretation

- Personal loans dominate the portfolio with \$334.6M in loan amount, representing approximately 76.8% of the portfolio.
- 32,716 loans (84.8%) are classified as fully recovered.
- Moderate risk is the largest risk segment at 30,363 loans (78.7%); High risk accounts for 7,873 loans (20.4%).
- Charged-off loans total 5,333, or 13.8% of the portfolio.
- Risk rises materially across grades: the average risk score moves from roughly 45.6% for Grade A to 77.7% for Grade G.

2. Report Structure

Section	Purpose
1	Executive Summary
2	Report Structure
3	Project Definition and Business Questions
4	Dataset and Source Structure
5	End-to-End Analytical Workflow
6	Excel Data Preparation
7	Power Query Cleaning and Feature Engineering
8	Data Quality Assessment
9	Power BI Data Model
10	Analytical Measures and DAX Framework
11	Dashboard Architecture
12	Dashboard Page 1 — Executive Overview
13	Dashboard Page 2 — Risk & Credit
14	Dashboard Page 3 — Profitability & Recovery
15	Dashboard Page 4 — Borrower Profile
16	Portfolio Performance Analysis
17	Loan Grade and Risk Analysis
18	Recovery and Profitability Analysis
19	Borrower and Income Analysis
20	Purpose and Geographic Analysis
21	SQL Independent Validation
22	Validation Findings and Controls
23	Key Business Insights
24	Recommendations
25	Limitations and Analytical Controls
26	Conclusion
27	Technical Skills Demonstrated
Appendix A	Data Dictionary
Appendix B	SQL Query Inventory
Appendix C	Evidence Register

3. Project Definition and Business Questions

The project is designed to turn loan-level financial records into a portfolio-monitoring and lending-decision analytical product. The work combines borrower profile, loan economics, recovery, profitability, credit risk and geographic analysis rather than focusing on a single KPI.

Business Questions

- How large is the loan portfolio in terms of loan count, amount and payments received?
- How effectively are issued loans being recovered, and how much exposure remains unrecovered?
- Which loan grades and risk categories carry the greatest credit exposure?
- How do DTI, LTI, EMI-to-income and interest-rate categories relate to risk?
- Which purposes and states contribute the greatest portfolio value?
- How does loan issuance and payment behavior change through the year?
- Which borrower income, employment and home-ownership groups dominate the portfolio?
- How do charged-off, loss and recovery categories affect portfolio economics?
- Can core portfolio metrics and analytical groupings be reproduced independently in SQL?

Project Objectives

- Prepare a consistent analytical table using Excel Power Query.
- Standardize source fields and create reusable business classifications.
- Build a dimensional Power BI model with a dedicated measures layer.
- Develop interactive executive, risk, profitability and borrower-profile pages.
- Use SQL to reproduce portfolio calculations and analytical summaries.
- Document data-quality conditions and controls instead of hiding exceptions.

4. Dataset and Source Structure

The supplied workbook contains a raw source worksheet named `Master_Data` and a 40-column analytical output named `Cleaned_Data`. The raw worksheet contains 38,576 records and 24 columns; the cleaned `Cleaned_Data` contains 38,576 records and 40 columns. Source database: Kaggle — Financial Loan Dataset (DataWithAryan). [Kaggle source](#)

Workbook object	Rows × columns	Role
Master_Data	38,576 × 24	Source-stage loan dataset
Cleaned_Data	38,576 × 40	Cleaned / engineered analytical dataset

Dataset Coverage

Dimension	Observed value
Loan records	38,576
Unique Loan_ID	38,576
Unique Customer_ID	38,576
Purpose categories	6
Purposes	14
States	50
Loan grades	7
Risk categories	3
Income groups	6
Loan issue period	01-Jan-2021 to 12-Dec-2021

Core Analytical Fields

Field group	Examples	Analytical role
Identifiers	Loan_ID, Customer_ID	Loan and borrower-level tracking
Loan economics	Loan_Amount, Total_Payment, EMI_Amount	Portfolio value and payment analysis
Credit	Loan_Grade, Loan_Sub_Grade, Risk_Score, Risk_Category	Credit-risk segmentation
Income / affordability	Annual_Income, DTI, LTI, EMI-to-Income	Borrower capacity analysis
Recovery	Loan_Recovered_Percent, Loan_Recovery_Category	Recovery performance
Status	Loan_Status, Profit_Status	Portfolio outcome classification
Time	Loan_Issue_Date, payment dates	Issuance and temporal analysis
Geography	State	State-level portfolio concentration

5. End-to-End Analytical Workflow

Stage	Tool	Primary activities	Output
1	Excel	Source workbook and raw loan records	Master_Data
2	Power Query	Cleaning, standardization, feature engineering and duplicate control	Cleaned Table / Cleaned_Data
3	Power BI	Fact/dimension modeling and measure layer	Analytical semantic model
4	Power BI	Interactive visuals, slicers and page-level analysis	4-page dashboard
5	SQL / MySQL	Independent KPI, risk, recovery, purpose, state and time analysis	Validation / analytical outputs
6	Business analysis	Interpretation, risk identification and recommendations	Decision-oriented conclusions

Traceability Principle

- Source-level transformations remain in Power Query rather than being manually overwritten in the analytical output.
- Derived business fields are created once and reused in downstream analysis.
- Power BI measures centralize recurring portfolio calculations.
- SQL provides an independent calculation path outside the Power BI visual layer.
- Dashboard screenshots are retained as evidence of the implemented reporting interface.

6. Excel Data Preparation

The workbook preserves a raw 24-column loan dataset and a 40-column analytical table. The analytical table retains 38,576 records and expands the field set through derived risk, recovery, profitability, affordability, term and validation classifications.

Layer	Columns	Key characteristics
Raw Master_Data	24	Original identifiers, dates, loan status, grade, borrower and financial fields
Cleaned Table / Cleaned_Data	40	Standardized fields plus derived analytical classifications and ratios

Source-to-Analytical Field Expansion

- Raw employment-length text is converted into a numeric employment-years field.
- Loan term text is converted into a numeric term measure and a term category.
- Purpose values are standardized and grouped into Purpose_Category.
- Income, loan size, interest rate, DTI and loan term are converted into analytical bands.
- Recovery percentage and recovery category are derived from loan amount and payment information.
- Profit status and risk measures are added for portfolio monitoring.
- A Data_Validation_Flag is added to identify records requiring date-quality attention.

7. Power Query Cleaning and Feature Engineering

The supplied PDF evidence shows a single 'Cleaned Table' query with a long, traceable Applied Steps sequence. The transformation is not limited to cosmetic cleanup; it also creates the analytical fields consumed by the Power BI report.

Step	Observed Applied Step
1	Source
2	Reordered Columns
3	Replaced null with unknown (Emp_Title)
4	Changed Data Type 1
5	Capitalized Each Word
6	Replaced Value in purpose
7	Replaced (years) with space in emp_length
8	Replaced Value2
9	Replaced <1 with 0 in emp_length
10	Replaced +10 with 10 in emp_length
11	Changed emp_length Type
12	Extracted No. from Term
13	Renamed Columns
14	Removed Constant Data Columns
15	Added Data_Validation_Flag
16	Added Income_Group
17	Added Loan_Size_Category
18	Added Employment_Level
19	Added Loan_to_Income_Ratio
20	Added EMI_to_Income_Ratio
21	Added Verified_Income_Flag
22	Added Loan_Recover_Percent
23	Changed Type2
24	Added Profit_Status
25	Added Risk_Aggregation
26	Added Interest_Rate_Category
27	Changed Type3
28	Added Risk_Score
29	Changed Type4
30	Added Risk_Category
31	Added Loan_Term_Category
32	Renamed Columns1
33	Added Loan_Recovery_Category
34	Added Purpose_Category
35	Added DTI_Category
36	Changed Type5
37	Reordered Columns1
38	Removed Duplicates

Power Query Evidence — 1

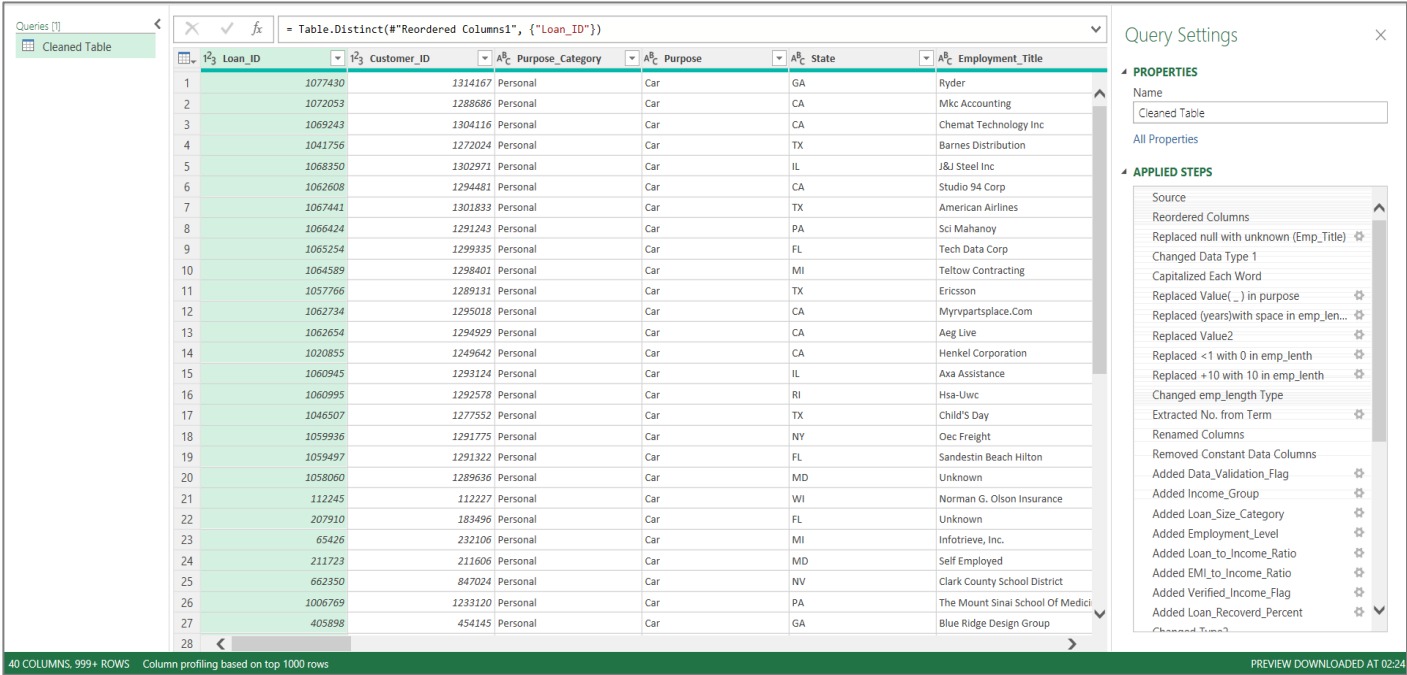


Figure 1. .1_Power_Query_Applied_Steps.png — Initial transformation and derived-field sequence.

Power Query Evidence — 2

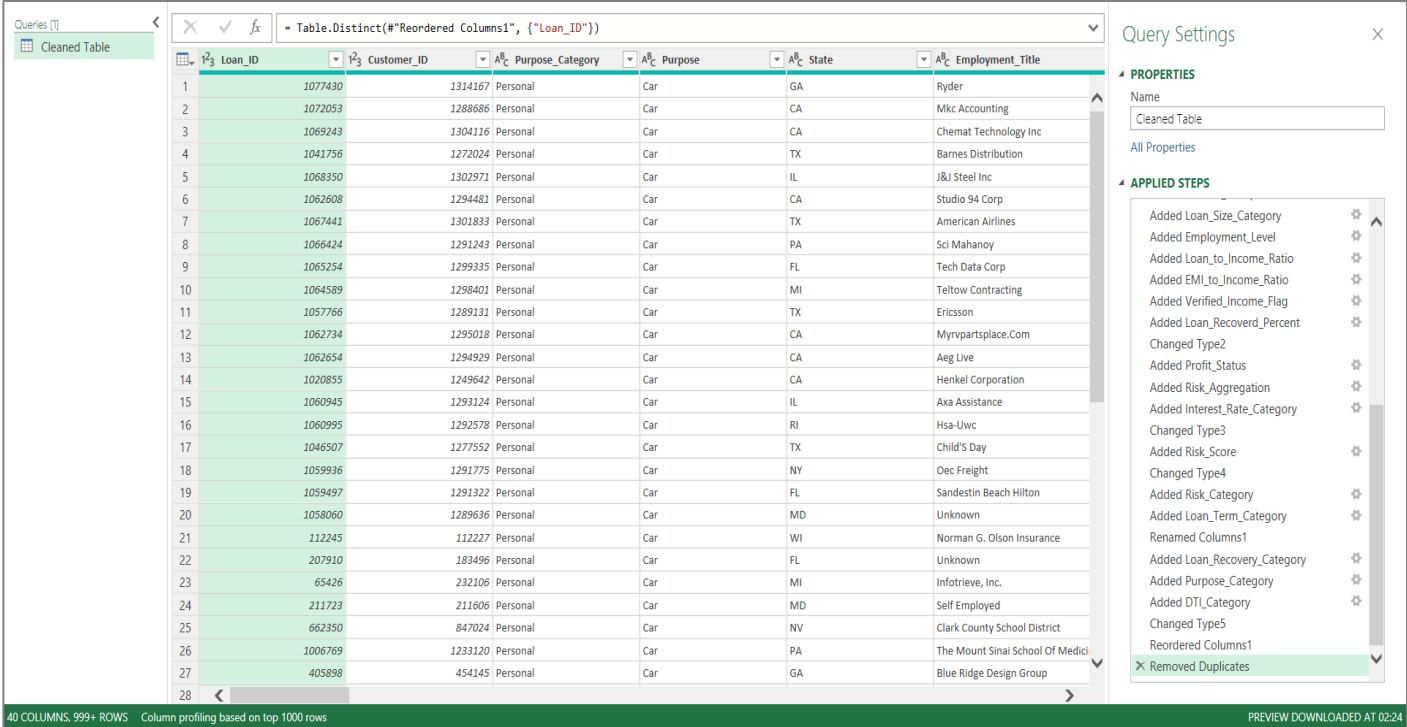


Figure 2. .2_Power_Query_Applied_Steps.png — Final classification and duplicate-removal steps.

8. Data Quality Assessment

Quality check	Observed result	Assessment
Total analytical rows	38,576	Complete row count in Cleaned_Data
Loan_ID duplicates	0	Loan_ID is unique in cleaned output
Customer_ID duplicates	0	Customer_ID is also one-to-one with Loan_ID in this dataset
Valid validation flag	23,123 (59.9%)	Dashboard reports approximately 60% valid-date quality
Invalid validation flag	15,453 (40.1%)	Large date-quality exception set
Missing Employment_Title	5	Power Query evidence shows nulls replaced with Unknown
Missing Loan_Amount	0	None
Missing Total_Payment	0	None

Validation Flag Diagnostic

The cleaned data contains 15,453 records flagged Invalid. A direct diagnostic shows that exactly 15,453 records have Last_Payment_Date earlier than Loan_Issue_Date, matching the invalid-record count. This strongly indicates that the validation flag is driven by this date-order condition. The report treats this as a source-data quality issue rather than silently rewriting the dates.

Data Quality Interpretation

- Loan_ID uniqueness is preserved after the duplicate-removal step.
- The validation flag materially reduces the usable 'valid' population to about 60% under the project's date-quality rule.
- Date-quality issues are especially important because loan timing affects repayment and portfolio-period interpretation.

9. Power BI Data Model

The supplied PBIX exposes a dimensional model containing a central Fact_Table, supporting borrower, date, loan and risk dimensions, and a dedicated _Measures table. The model structure separates transaction-level financial data from descriptive attributes and reusable calculations.

Model object	Observed role
Fact_Table	Central analytical loan-level fact table
Dim_Customer	Borrower profile attributes such as income, employment, ownership and state
Dim_Date	Date and month analysis
Dim_Loan	Loan attributes such as purpose, grade, term, status and recovery
Dim_Risk	Risk classification attributes
_Measures	Dedicated measure layer used by dashboard visuals

Modeling Principles

- Keep loan-level numeric measures in the central fact table.
- Use dedicated dimensions for borrower, loan, date and risk slicing.
- Centralize recurring KPIs in _Measures.
- Use slicers to drive cross-page portfolio segmentation.
- Maintain a distinction between source attributes and analytical classifications.

10. Analytical Measures and DAX Framework

The PBIX contains a dedicated _Measures layer. The report metadata exposes a reusable measure inventory covering portfolio KPIs, risk, recovery, profitability, borrower profile and data-quality monitoring.

Measure group	Observed measures
Portfolio	Total Loan Count, Total Loan Amount, Total Payments Received, Total Interest, Total Profit
Profitability / recovery	Profit Margin %, Recovery %, Unrecovered Amount, Profitable Loans Count, Loss Loans Count
Risk / credit	High Risk Loans Count, High Risk Count %, Moderate Risk Loans Count, Avg Risk Score, Charged Off Loan Count
Borrower	Total Customers, Avg Annual Income, Avg Loan Term, Avg employment Length, Avg Credit A/C, Verified Borrowers
Affordability	Avg DTI, Avg LTI, Avg EMI to Income
Data quality / refresh	In-Valid Loans, Last_Refresh

The supplied PBIX package exposes measure names and visual queries, but not every internal DAX expression as plain text. Therefore, this report documents the observed measure architecture without inventing unavailable formula bodies.

11. Power BI Dashboard Architecture

Page	Purpose	Major components observed
Overview	Executive portfolio monitoring	KPI cards, monthly loan/payment trend, grade, purpose, state, status, risk and DTI views
Risk & Credit Analysis	Credit-risk monitoring	KPI cards, risk distribution, risk by grade, charged-off analysis, DTI, purpose and interest-rate risk
Profitability & Recovery Analysis	Economic performance	KPI cards, monthly profit, profitability by grade/purpose, recovery by grade, profit status and recovery categories
Borrower Profile	Borrower composition	KPI cards, income, DTI/LTI, purpose, state, home ownership, employment and average loan views

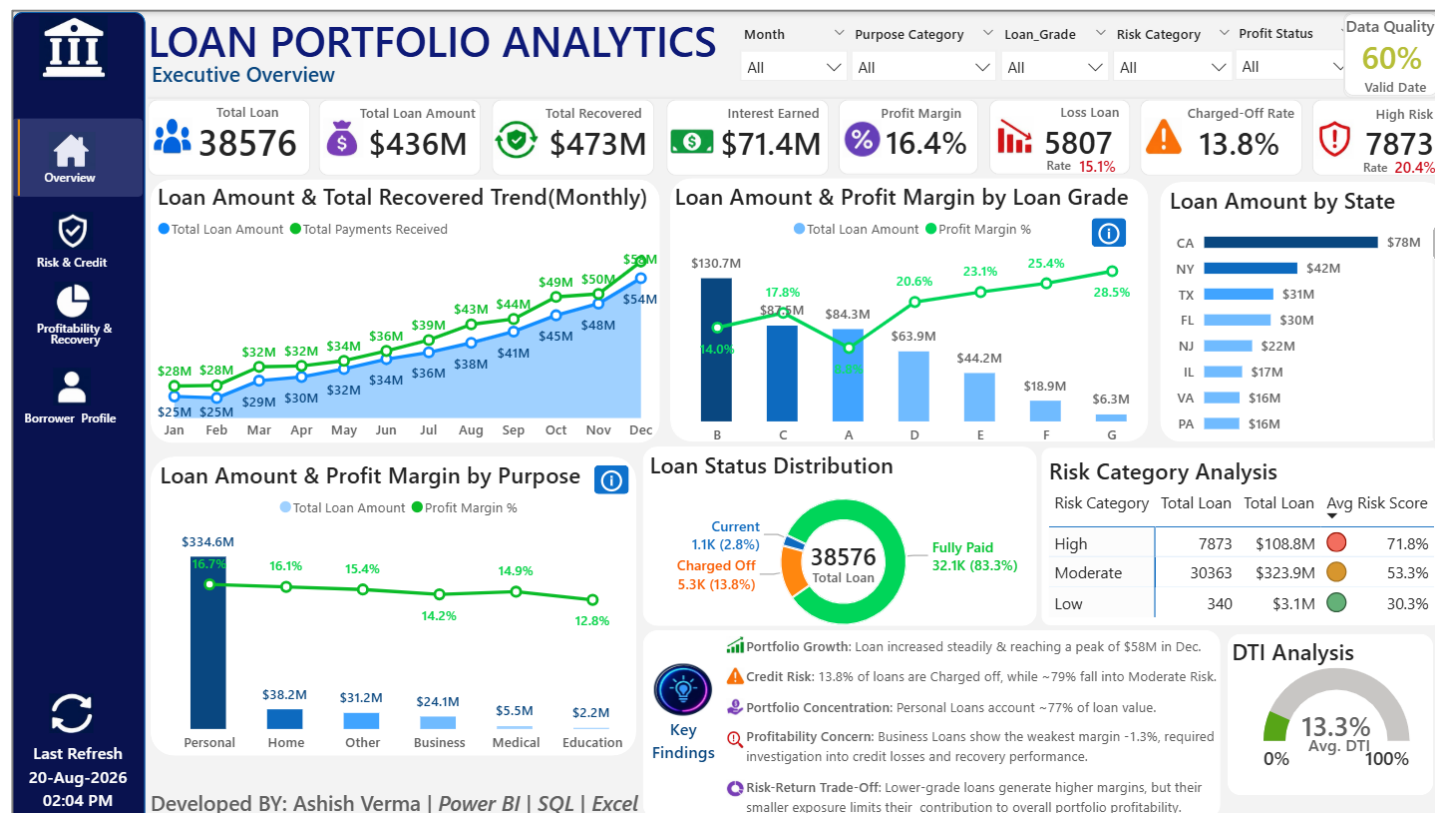
Common Interaction Layer

- Month
- Purpose Category
- Loan Grade
- Risk Category
- Profit Status

The dashboard uses a consistent navigation pattern and common slicer logic to move from executive portfolio position into risk, profitability/recovery and borrower-level interpretation.

12. Dashboard Page 1 — Executive Overview

The Executive Overview is the primary management page. It establishes portfolio scale, payment behavior, profitability, risk exposure, state concentration and status distribution in a single view.



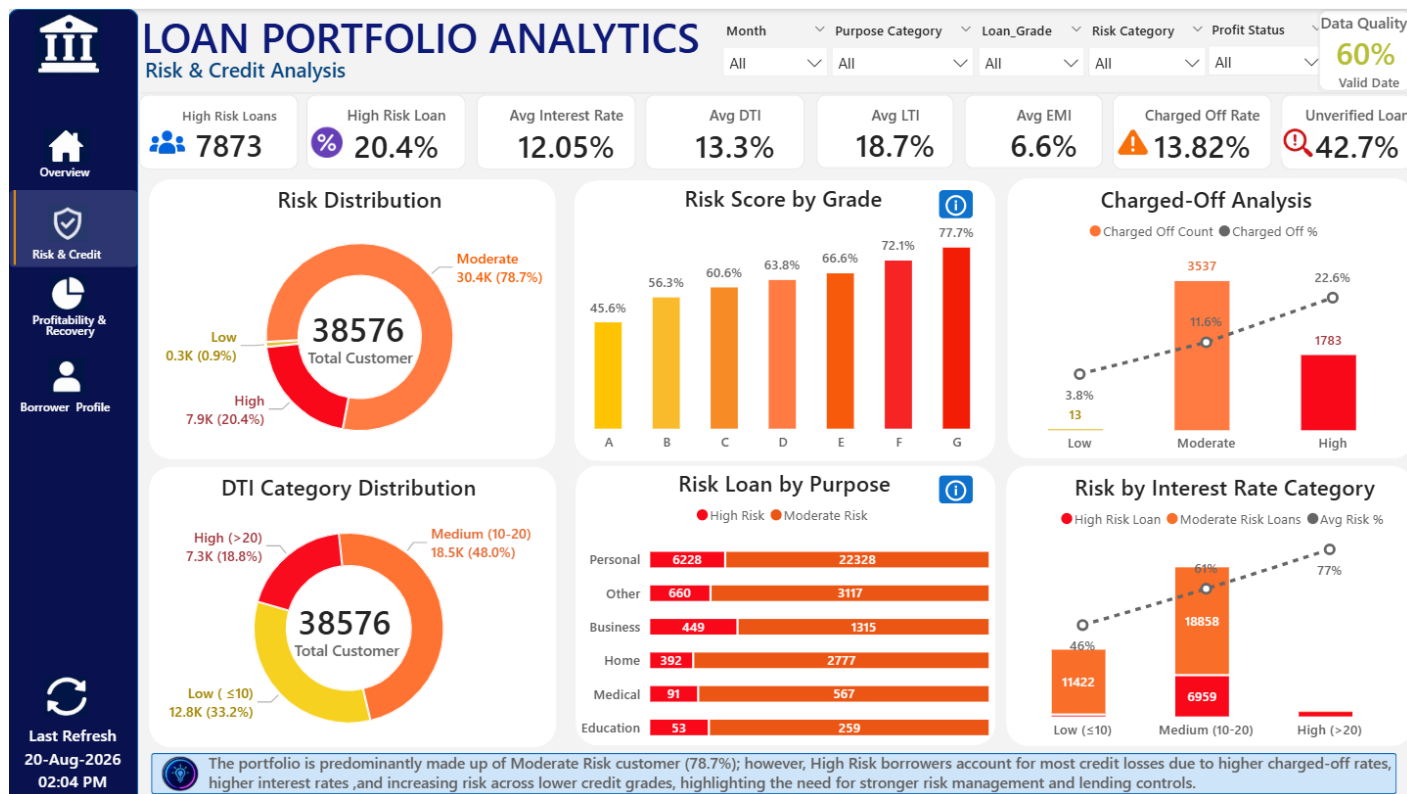
Dashboard evidence — 01_Dashboard_Overview.png.

Analytical Observations

- Total loan count shown is 38,576; total loan amount is approximately \$435.8M.
- Total payments received are approximately \$473.1M, above total loan amount on an aggregate basis.
- Personal purpose loans account for about 77% of portfolio value.
- Grade B carries the largest loan amount at \$130.7M.
- California is the largest state by loan amount at \$78.5M.

13. Dashboard Page 2 — Risk & Credit

The Risk & Credit page focuses on exposure, risk scoring, DTI, charged-off behavior and the relationship between loan grade, interest-rate category and risk.



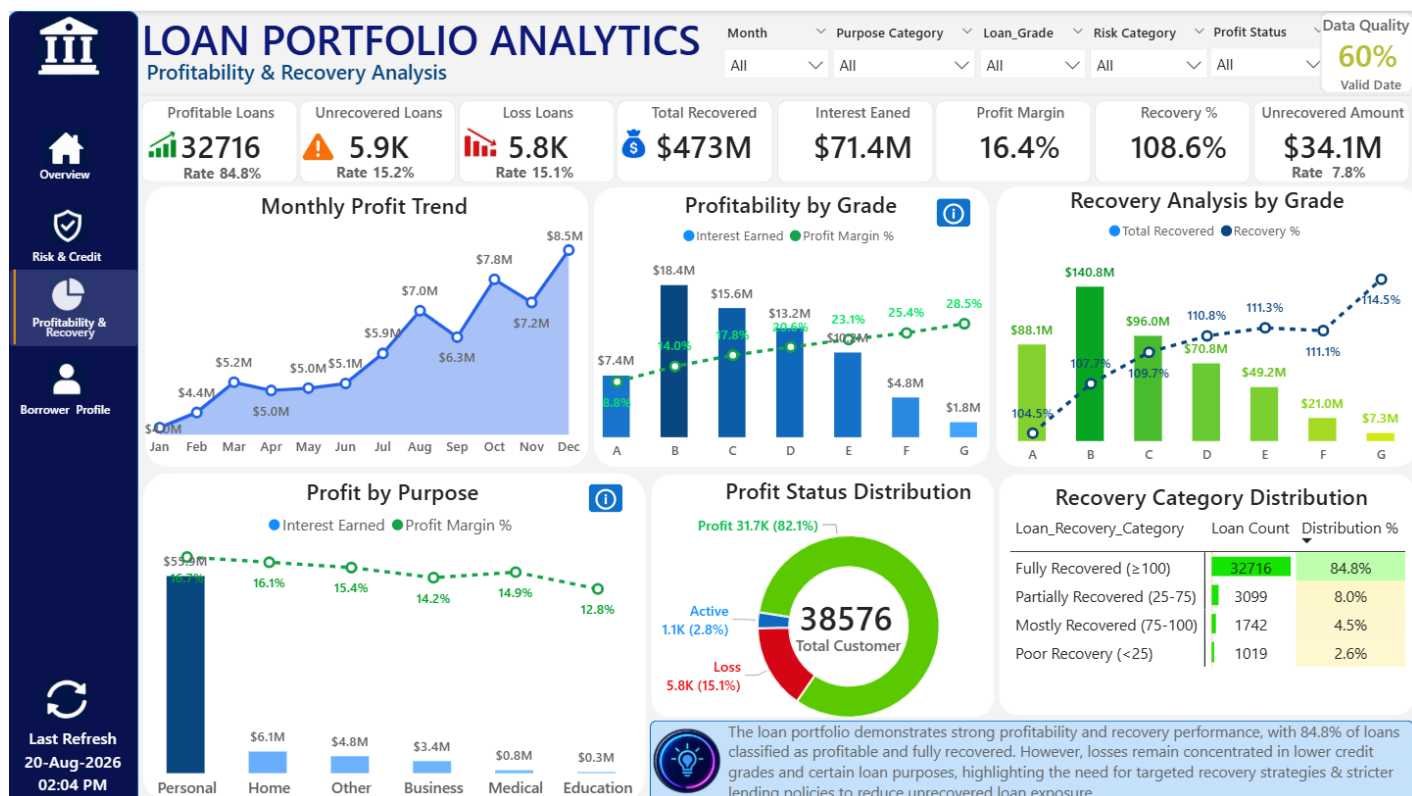
Dashboard evidence — 02_Dashboard_Risk_Credit_Analysis.png.

Analytical Observations

- Moderate risk represents 30,363 loans (78.7%).
- High-risk loans total 7,873 (20.4%).
- Average DTI is 13.3%, while average LTI is 18.7%.
- Charged-off loans total 5,333 (13.8%).
- Average risk score increases from roughly 45.6% in Grade A to 77.7% in Grade G.

14. Dashboard Page 3 — Profitability & Recovery

The Profitability & Recovery page separates economic return from credit-risk exposure. It shows monthly profit, grade economics, recovery performance, purpose profitability and profit-status composition.



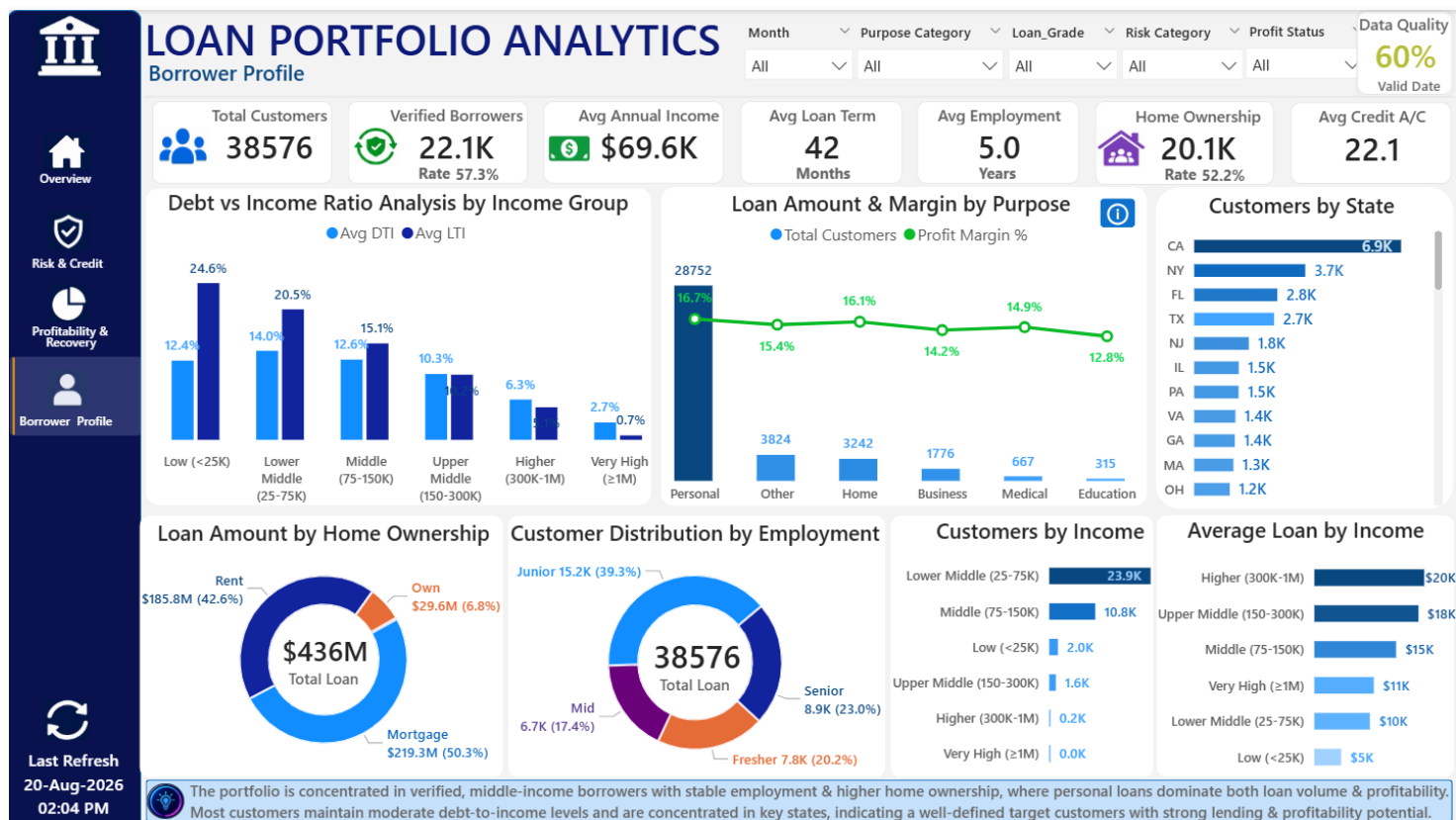
Dashboard evidence — 03_Dashboard_Profitability_Recovery_Analysis.png.

Analytical Observations

- Fully recovered loans total 32,716, or 84.8%.
- Unrecovered / non-fully-recovered loans total 5,860.
- Interest earned under the project definition is \$71.4M, producing a 16.4% margin measure.
- Grade G shows the highest margin at approximately 28.5%, while Grade A is approximately 8.8%.
- The dashboard shows loss loans at 5,807 (15.1%).

15. Dashboard Page 4 — Borrower Profile

The Borrower Profile page explains who the portfolio serves. It combines borrower income, affordability ratios, employment, home ownership, purpose and state concentration.



Dashboard evidence — 04_Dashboard_Borrower_Analysis.png.

Analytical Observations

- Average annual income is approximately \$70K.
- Verified-income records total 22,112 (57.3%).
- Average loan term is approximately 42.4 months and average employment length is 5.0 years.
- Lower Middle income is the largest income group with 23,908 records.
- Mortgage and Rent dominate home-ownership categories; non-rent ownership/mortgage records represent about 52.2%.

16. Portfolio Performance Analysis

The source data covers calendar year 2021, with monthly loan issuance increasing through the year. December is the largest issuance month by loan count and loan amount.

Month	Loans	Loan Amount	Payments	Recovery	Margin
Jan	2,332	\$25.0M	\$27.6M	110.2%	16.1%
Feb	2,279	\$24.6M	\$27.7M	112.5%	17.9%
Mar	2,627	\$28.9M	\$32.3M	111.7%	17.9%
Apr	2,755	\$29.8M	\$32.5M	109.0%	16.7%
May	2,911	\$31.7M	\$33.8M	106.3%	15.8%
Jun	3,184	\$34.2M	\$36.2M	105.9%	15.0%
Jul	3,366	\$35.8M	\$38.8M	108.4%	16.5%
Aug	3,441	\$38.1M	\$42.7M	111.9%	18.3%
Sep	3,536	\$40.9M	\$44.0M	107.5%	15.4%
Oct	3,796	\$44.9M	\$49.4M	110.0%	17.3%
Nov	4,035	\$47.8M	\$50.1M	105.0%	15.1%
Dec	4,314	\$54.0M	\$58.1M	107.6%	15.8%

December contains 4,314 loans and \$54.0M in issuance, the highest monthly level. January starts at 2,332 loans and \$25.0M. The trend is therefore strongly upward in issuance volume across the year.

	Month_No	Month_	Total_Loans	Total_Loan_Amount	Total_Recovered_Amount	Interest_Received	Loss_Amount	Profit_Margin	Unrecovered_Loan_Amount
▶	1	Jan	2332	25031650	27578836	4022524	1475338	16.07	1475338
	2	Feb	2279	24647825	27717745	4405643	1335723	17.87	1335723
	3	Mar	2627	28875700	32264400	5171027	1782327	17.91	1782327
	4	Apr	2755	29800800	32495533	4962951	2268218	16.65	2268218
	5	May	2911	31738350	33750523	5021611	3004320	15.82	3009438
	6	Jun	3184	34161475	36164533	5139611	3114554	15.05	3136553
	7	Jul	3366	35813900	38827220	5897757	2878404	16.47	2884437
	8	Aug	3441	38149600	42682218	6978970	2445935	18.29	2446352
	9	Sep	3536	40907725	43983948	6310456	3174531	15.43	3234233
	10	Oct	3796	44893800	49399567	7771773	3266006	17.31	3266006
	11	Nov	4035	47754825	50132030	7188255	4811050	15.05	4811050
	12	Dec	4314	53981425	58074380	8511730	4348743	15.77	4418775

Result 4 ×

Output

Action Output

#	Time	Action	Message
✔ 4	01:39:43	WITH Time_Analysis AS (SELECT *, MONTH(Loan_Issue_Date)AS Month_No, DATE_FO...	12 row(s) returned

Independent SQL Validation – 05_ SQL_Validation.sql

17. Loan Grade and Risk Analysis

Grade	Loans	Loan Amount	Interest	Margin	Charged-Off Rate	Avg Risk Score
A	9,689	\$84.3M	\$7.4M	8.8%	5.7%	45.6%
B	11,674	\$130.7M	\$18.4M	14.0%	11.5%	56.3%
C	7,904	\$87.5M	\$15.6M	17.8%	16.0%	60.6%
D	5,182	\$63.9M	\$13.2M	20.6%	20.7%	63.8%
E	2,786	\$44.2M	\$10.2M	23.1%	24.8%	66.6%
F	1,028	\$18.9M	\$4.8M	25.4%	30.3%	72.1%
G	313	\$6.3M	\$1.8M	28.5%	31.3%	77.7%

Loan grade shows a clear risk-return progression. Higher grades carry progressively higher average risk scores and charged-off rates, while the project-defined margin also rises with grade. This is an important portfolio trade-off: higher-risk grades may generate stronger interest economics but require tighter credit controls.

	Loan_Grade	Total_Loan	Total_Loan_Amount	Interest_Received	Profit_Margin	Total_Recovered_Amount	Loss_Amount	Unrecovered_Loan_Amount	Avg_Risk_Score
▶	A	9689	84252225	7439407	8.83	88051563	3640069	3640069	4.56
	B	11674	130703975	18356058	14.04	140775015	8187522	8285018	5.63
	C	7904	87456450	15585556	17.82	95973518	7049718	7068488	6.06
	D	5182	63920800	13168374	20.60	70823891	6250626	6265283	6.38
	E	2786	44165100	10223264	23.15	49164151	5195851	5224213	6.66
	F	1028	18910450	4800175	25.38	21016738	2689871	2693887	7.21
	G	313	6348075	1809474	28.50	7266057	891492	891492	7.77

Result 8 x

Output

Action Output

#	Time	Action	Message
8	01:46:45	SELECT Loan_Grade, COUNT(Loan_ID) AS Total_Loan, SUM(Loan_Amount)AS Total_Loan_Amount, ...	7 row(s) returned

Independent loan volume and average risk score by grade–06_ SQL_Validation.png

Risk Category

Risk	Loans	Loan Amount	Portfolio Share	Charged-Off Rate	Avg Risk Score
Low	340	\$3.1M	0.9%	3.8%	30.3%
Moderate	30,363	\$323.9M	78.7%	11.6%	53.3%
High	7,873	\$108.8M	20.4%	22.6%	71.8%

	Risk_Category	Total_Loan	Loan_Distribution	Total_Loan_Amount	Charged_Off	Charged_Off_Rate	Profit_Margin
▶	Moderate	30363	78.71	323866200	3537	11.65	15.08
	High	7873	20.41	108821775	1783	22.65	20.49
	Low	340	0.88	3069100	13	3.82	8.33

Result 10 x

Output

Action Output

#	Time	Action	Message
10	01:47:54	SELECT Risk_Category, COUNT(Loan_ID) AS Total_Loan, ROUND(COUNT(Loan_ID)*100/	(S... 3 row(s) returned

Independent loan risk category–07_ SQL_Validation.png

High-risk exposure is material: 7,873 loans account for roughly one-fifth of the portfolio. The high-risk group also carries the highest average risk score and should therefore be monitored separately from the much larger Moderate-risk population.

18. Recovery and Profitability Analysis

Recovery Category	Loans	Share	Loan Amount	Payments
Fully Recovered (≥ 100)	32,716	84.8%	\$356.3M	\$427.7M
Mostly Recovered (75-100)	1,742	4.5%	\$25.8M	\$22.3M
Partially Recovered (25-75)	3,099	8.0%	\$40.8M	\$21.0M
Poor Recovery (< 25)	1,019	2.6%	\$12.9M	\$2.0M

Profit Status

Profit Status	Loans	Share	Loan Amount	Payments
Profit	31,671	82.1%	\$338.8M	\$404.7M
Loss	5,807	15.1%	\$78.1M	\$44.1M
Active	1,098	2.8%	\$18.9M	\$24.2M

The recovery structure is favorable in count terms, with 84.8% of loans classified as fully recovered. However, the remaining 15.2% of loans span mostly, partially and poorly recovered categories and represent approximately \$34.1M of unrecovered exposure under the project's calculation.

19. Borrower and Income Analysis

Income Group	Loans	Portfolio Share	Avg DTI	Avg LTI
Low (<25K)	2,046	5.3%	12.4%	24.6%
Lower Middle (25-75K)	23,908	62.0%	14.0%	20.5%
Middle (75-150K)	10,849	28.1%	12.6%	15.1%
Upper Middle (150-300K)	1,553	4.0%	10.3%	10.2%
Higher (300K-1M)	206	0.5%	6.3%	5.1%
Very High (≥1M)	14	0.0%	2.7%	0.7%

Employment Level

Employment Level	Loans	Share	Avg Income	Avg Employment Years
Fresher	7,804	20.2%	\$62K	0.4
Junior	15,171	39.3%	\$66K	3.4
Mid	6,731	17.4%	\$71K	7.3
Senior	8,870	23.0%	\$82K	10.0

Home Ownership

Home Ownership	Loans	Portfolio Share	Loan Amount
Rent	18,439	47.8%	\$185.8M
Mortgage	17,198	44.6%	\$219.3M
Own	2,838	7.4%	\$29.6M
Other	98	0.3%	\$1.0M

The borrower population is concentrated in the Lower Middle income group and Junior employment level. The affordability fields show meaningful variation across income groups, supporting the use of DTI/LTI segmentation alongside simple income ranking.

20. Purpose and Geographic Analysis

Purpose Category	Loans	Loan Amount	Margin	Charged-Off Rate
Personal	28,752	\$334.6M	16.7%	13.1%
Home	3,242	\$38.2M	16.1%	11.8%
Other	3,824	\$31.2M	15.4%	15.4%
Business	1,776	\$24.1M	14.2%	25.6%
Medical	667	\$5.5M	14.9%	15.0%
Education	315	\$2.2M	12.8%	15.9%

Personal loans are the dominant portfolio driver. Business loans are smaller in value but show the highest charged-off rate among the major purpose categories, making them a useful risk-monitoring segment.

Top States by Loan Amount

State	Loans	Loan Amount	Payments	Margin
CA	6,894	\$78.5M	\$83.9M	15.7%
NY	3,701	\$42.1M	\$46.1M	16.5%
TX	2,664	\$31.2M	\$34.4M	16.7%
FL	2,773	\$30.0M	\$31.6M	15.5%
NJ	1,822	\$21.7M	\$23.4M	16.4%
IL	1,486	\$17.1M	\$18.9M	16.9%
VA	1,375	\$16.0M	\$17.7M	17.2%
PA	1,482	\$15.8M	\$17.5M	16.7%
GA	1,355	\$15.5M	\$16.7M	16.4%
MA	1,310	\$15.1M	\$16.7M	16.8%

California is the largest state by portfolio amount, followed by New York, Texas and Florida. Geographic concentration should be evaluated alongside risk and recovery rather than by loan amount alone.

21. SQL Independent Validation

The supplied SQL script is written for MySQL 8.0+ and explicitly treats the project as a single-table analysis, so JOIN operations are not required. The script is organized into database setup, data import, quality validation, KPI analysis, recovery, borrower profile, time, grade/risk, geography, purpose and final portfolio analysis.

SQL group	Purpose
01. Database & Table Setup	Create Finance_DB and finance_data schema
02. Data Import & MySQL Configuration	Enable local infile and load cleaned CSV
03. Data Structure & Quality Validation	Preview, count, duplicate and missing-field checks
04. Overall Loan Portfolio KPI Analysis	Executive, risk, profitability/recovery and borrower KPIs
05. Loan Recovery & Profitability Analysis	Recovery category, loan status and profit status
06. Customer & Borrower Profile Analysis	Income groups and employment levels
07. Time & Loan Issuance Analysis	Monthly issuance, payments, profit and recovery
08. Loan Grade & Risk Analysis	Grade, risk category and DTI
09. Geographic Analysis	State-level portfolio performance
10. Loan Purpose Analysis	Purpose categories and top loss-making purposes
12. Final Financial Portfolio Analysis	Final consolidated portfolio metrics

The source script jumps from section 10 to section 12; no section 11 is present. The report preserves that observation rather than silently renumbering the source artifact.

22. Validation Findings and Controls

Core Portfolio Reconciliation

Metric	Cleaned-data calculation	Dashboard evidence	SQL definition / equivalent
Loan count	38,576	38,576	COUNT(Loan_ID)
Loan amount	\$435.8M	\$436M	SUM(Loan_Amount)
Payments received	\$473.1M	\$473M	SUM(Total_Payment)
Interest earned	\$71.4M	\$71.4M	Fully recovered: Total_Payment – Loan_Amount
Profit margin	16.4%	16.4%	Interest ÷ total loan amount
Recovery rate	108.6%	108.6%	SUM(Total_Payment) ÷ SUM(Loan_Amount)
Unrecovered amount	\$34.1M	\$34.1M	Non-fully-recovered Loan_Amount – Total_Payment
Charged-off rate	13.8%	13.8%	Charged Off count ÷ total loans
High-risk loans	7,873	7,873	Risk_Category = High

Important Methodological Controls

- Customer_ID is unique in the supplied cleaned table and has the same cardinality as Loan_ID. Therefore, dashboard labels such as 'Total Customers' should be interpreted carefully: 38,576 is effectively the number of loan/borrower records, not evidence of multiple loans per borrower.
- The dashboard's Data Quality indicator is approximately 60%, matching 23,123 Valid records out of 38,576.
- The 15,453 Invalid records exactly match the count of records where Last_Payment_Date precedes Loan_Issue_Date.
- The SQL script scales Risk_Score by ×10 in its grade and purpose outputs, while the dashboard intentionally presents the underlying 0-1 risk score as a percentage(×100) to represent the estimate chance of risk. Therefore, the difference in scale is intentional and is used for clear dashboard interpretation.
- The SQL query label 'Loan_Distribution' divides loan counts by distinct Customer_ID. This happens to align here because Customer_ID is one-to-one with Loan_ID, but COUNT(DISTINCT Loan_ID) would be a clearer denominator if the model later permits multiple loans per customer.

23. Key Business Insights

- 1. Portfolio scale:** 38,576 loans represent approximately \$435.8M of issued loan value.
- 2. Recovery strength:** 32,716 loans (84.8%) are fully recovered.
- 3. Unrecovered exposure:** non-fully-recovered loans leave approximately \$34.1M unrecovered(7.8%) under the project's category-based calculation.
- 4. Risk concentration:** Moderate risk dominates at 30,363 loans, but High risk contributes 7,873 loans.
- 5. Credit deterioration by grade:** charged-off rate rises from 5.7% in Grade A to 31.3% in Grade G.
- 6. Grade economics:** margin rises from 8.8% in Grade A to 28.5% in Grade G under the project definition.
- 7. Purpose concentration:** Personal loans account for roughly 77% of portfolio amount.
- 8. Business-purpose risk:** Business loans have a charged-off rate of 25.6%, higher than the other major purpose categories.
- 9. Geographic concentration:** California is the largest state by loan amount at \$78.5M.
- 10. Borrower profile:** Lower Middle income is the largest income group, while Junior employment level is the largest employment classification.
- 11. Temporal concentration:** December is the strongest issuance month, with 4,314 loans and \$54.0M issued.
- 12. Data quality:** approximately 40% of records are flagged Invalid under the project's validation flag, driven by payment-before-issue date ordering.

24. Recommendations

Recommendation	Action	Priority
Strengthen high-risk controls	Use Risk Category, grade, DTI and charged-off behavior together for underwriting and monitoring.	High
Target unrecovered exposure	Prioritize Mostly, Partially and Poor Recovery groups for collection and recovery strategy review.	High
Review Business-purpose lending	Business loans show a comparatively high charged-off rate; evaluate underwriting, pricing and recovery separately.	High
Use grade-based pricing controls	Higher-risk grades generate higher margin under the project definition but also show materially higher charged-off rates.	High
Address date-quality exceptions	Investigate the 15,453 records with payment dates preceding issue dates before relying on time-based repayment analysis.	High
Improve metric naming	Rename or clarify 'Total Customers' where the metric is actually a loan/borrower-record count.	Medium
Validate risk-score scaling	Correct the SQL $\times 10$ risk-score scaling to $\times 100$ if percentage reconciliation is intended.	High
Monitor state concentration	Combine state-level loan amount with risk, recovery and charged-off rate to avoid volume-only decisions.	Medium
Segment borrower affordability	Use income group, DTI, LTI and EMI-to-income together rather than annual income alone.	Medium
Retain SQL QA layer	Re-run the SQL validation suite whenever Power Query or Power BI model logic changes.	High

25. Limitations and Analytical Controls

Known Limitations

- The supplied data is a 2021 loan snapshot; it should not be interpreted as a current live lending portfolio.
- The source contains substantial date-order anomalies captured by the validation flag.
- Customer_ID is unique per loan in this dataset, limiting the ability to study repeat-borrower behavior.
- Profit margin is a project-defined metric based on interest from fully recovered loans divided by total loan amount; it should not automatically be treated as accounting net profit.
- Recovery above 100% at aggregate level reflects total payments exceeding total issued loan amount and should be interpreted as cumulative payment recovery, not a conventional recovery ratio constrained to 100%.
- The PBIX exposes measure names and visual structures, but not every internal DAX expression in plain text.

Controls Strengthening Reliability

- Power Query provides a repeatable transformation chain.
- Loan_ID uniqueness is explicitly checked.
- Missing critical numeric fields are checked.
- Date validation is surfaced through Data_Validation_Flag.
- SQL independently reproduces core totals and category-level analyses.
- Dashboard evidence is retained as page-level documentation.

26. Conclusion

The Financial Loan Portfolio Analysis project establishes a complete analytical workflow from raw loan records to a cleaned 40-column analytical dataset, Power BI semantic model and four-page interactive dashboard, supported by an independent MySQL SQL analysis layer. The portfolio contains 38,576 loan records and approximately \$435.8M in loan amount.

The business picture is characterized by strong aggregate payment recovery, substantial fully recovered loan volume, a dominant Moderate-risk population and meaningful High-risk exposure. Personal lending drives portfolio scale, while Business-purpose loans warrant closer credit monitoring. Risk scores and charged-off rates rise across lower loan grades, highlighting a clear risk-return trade-off.

The project's strongest analytical feature is the separation of responsibilities between tools: Excel and Power Query prepare the analytical dataset, Power BI organizes the semantic model and interactive decision views, and SQL provides an independent validation and analysis layer. The main improvement opportunity is to resolve the documented date-quality exceptions and the SQL risk-score scaling inconsistency before treating every metric as fully reconciled.

27. Technical Skills Demonstrated

Skill area	Evidence in project
Excel data handling	Raw and cleaned workbook layers; structured analytical table
Power Query	38 observed applied steps, data typing, text cleanup, standardization and derived fields
Data quality	Validation flag, missing-value checks, duplicate checks and date diagnostics
Data modeling	Fact_Table plus borrower, date, loan and risk dimensions
DAX / measures	Dedicated _Measures layer with portfolio, risk, recovery and borrower metrics
Dashboard design	Four analytical pages with KPI cards, slicers, charts, matrices, status and geographic views
SQL / MySQL	Database setup, import, quality validation, aggregations, CTE, window-function ranking
Financial analysis	Loan amount, payments, interest, recovery, profitability and affordability ratios
Credit-risk analysis	Grade, risk category, DTI, interest-rate and charged-off analysis
Business communication	Executive findings, risk observations and action-oriented recommendations
Validation	Independent SQL calculations and source-to-dashboard control checks

Appendix A — Data Dictionary

Field	Definition	Role
Loan_ID	Loan identifier	Identifier
Customer_ID	Borrower identifier	Identifier
Purpose_Category	Grouped loan purpose	Category
Purpose	Specific loan purpose	Category
State	Borrower state	Geography
Employment_Title	Employment title	Text
Employment_Level	Fresher / Junior / Mid / Senior classification	Derived
Emp_Length_Years	Numeric employment length	Numeric
Home_Ownership	Home ownership category	Category
Annual_Income	Annual borrower income	Numeric
Income_Group	Income band	Derived
Income_Verification_Status	Income verification state	Category
Verified_Income_Flag	Binary income verification flag	Derived
Debt_to_Income_Ratio	Debt relative to income	Numeric
DTI_Category	DTI band	Derived
Total_Credit_Account	Total credit accounts	Numeric
Loan_Amount	Original loan amount	Numeric
Loan_Size_Category	Loan amount band	Derived
Interest_Rate	Loan interest rate	Numeric
Interest_Rate_Category	Interest-rate band	Derived
EMI_Amount	Installment amount	Numeric
Loan_Term_Months	Loan term in months	Numeric
Loan_Term_Category	Short / Long term	Derived
Loan_Grade	Loan credit grade	Category
Loan_Sub_Grade	Loan sub-grade	Category
Loan_Issue_Date	Loan issue date	Date
Last_Credit_Check_Date	Last credit-check date	Date
Last_Payment_Date	Last payment date	Date
Next_Payment_Due_Date	Next scheduled payment date	Date
Loan_to_Income_Ratio	Loan amount relative to income	Derived
EMI_to_Income_Ratio	EMI relative to income	Derived
Total_Payment	Aggregate payment amount	Numeric
Loan_Recovered_Percent	Payment recovery percentage	Derived
Loan_Recovery_Category	Recovery classification	Derived
Loan_Status	Current / paid / charged-off status	Category
Profit_Status	Profit / loss / active classification	Derived
Risk_Score	Calculated risk score	Derived
Risk_Category	Low / Moderate / High	Derived
Risk_Aggregation	Risk aggregation measure	Derived
Data_Validation_Flag	Record validation result	Derived

Appendix B — SQL Query Inventory

Group	Queries / scope	Analytical purpose
01	CREATE DATABASE, USE, CREATE TABLE	Establish Finance_DB and finance_data
02	SET GLOBAL local_infile, SHOW VARIABLES, LOAD DATA	Configure and import data
03	Preview, row count, duplicate Loan_ID, missing critical fields	Structural validation
04.01	Executive overview KPIs	Loan count, amount, payments, interest, margin, losses, charge-offs, high risk
04.02	Risk and category KPIs	Risk, interest, DTI, LTI, EMI-to-income, charge-offs, verification
04.03	Recovery and profitability KPIs	Profitable/unrecovered loans, interest, margin, recovery
04.04	Borrower profile KPIs	Customers, verification, income, term, employment, ownership, credit
05.01	Recovery category performance	Recovery distribution and unrecovered amount
05.02	Loan status performance	Fully Paid / Charged Off / Current
05.03	Profit status performance	Profit / Loss / Active
06.01	Income group analysis	Borrower distribution and affordability
06.02	Employment level analysis	Borrower distribution, income and employment
07.01	Monthly issuance analysis	Monthly loans, amount, payments, interest, loss and recovery
08.01	Loan grade performance	Grade-level amount, interest, margin, recovery, loss and risk
08.02	Risk category performance	Risk distribution, charged-off rate and margin
08.03	DTI category analysis	DTI distribution and charged-off behavior
09.01	State-level portfolio performance	Geographic concentration and profitability
10.01	Purpose category performance	Purpose-level portfolio and risk economics
10.02	Top 3 loss-making purposes	Window-function ranking within purpose category
12	Final financial portfolio analysis	Consolidated portfolio KPI output

The supplied SQL also includes CTE and ROW_NUMBER() logic for ranking loss-making purposes within each Purpose_Category. The query set is intentionally single-table and does not require JOINS.

Appendix C — Evidence Register

Artifact	Used for
Finance_Loan_Portfolio_Cleaned_Data.xlsx	Raw source data, cleaned Cleaned_Data output, workbook structure and field values
Finance_Loan_Portfolio_Analysis.pbix	Power BI page names, visual query metadata, model objects and _Measures inventory
Finance_Loan_Portfolio_SQL_Validation.sql	MySQL schema, validation logic, portfolio analysis and ranking queries
Dashboard & Power Query screenshots	01_Dashboard_Overview.png; 02_Dashboard_Risk_Credit_Analysis.png; 03_Dashboard_Profitability_Recovery_Analysis.png; 04_Dashboard_Borrower_Analysis.png; .1_Power_Query_Applied_Steps.png; .2_Power_Query_Applied_Steps.png — visual evidence of the implemented dashboard and Power Query steps.
Independent calculated charts	Monthly portfolio trend and grade/risk analytical views generated from supplied cleaned data
Kaggle database source	Financial Loan Dataset — DataWithAryan (source database): https://www.kaggle.com/datasets/datawitharyan/financial-loan-dataset

Final Analytical Deliverable

This report is intended to accompany Finance_Loan_Portfolio_Analysis.pbix.pbix, Master_Data_Portfolio_Cleaned_Data.xlsx.xlsx and Finance_Loan_Portfolio_SQL_Validation.sql.sql as a single portfolio-ready financial analytics package. The report documents the implementation sequence, analytical outputs, business interpretation, controls and limitations without inventing unavailable source details.